

Sarvam AI TTS Dataset Report

1. Project Overview

I built a complete Text-to-Speech (TTS) training dataset for Indian English (en-IN) and Hindi (hi-IN) using YouTube-sourced clean, single-speaker educational and lecture audio clips. The pipeline includes automatic audio downloading, audio format conversion, silence-based segmentation, transcription via Sarvam ASR, emotion/style tagging using Sarvam LLM, automated quality filtering, manual quality control, and packaging as a HuggingFace-compatible dataset.

2. Dataset Composition

The final processed and validated dataset has the following statistics:

- **Accepted clips:** 125
- **Total duration:** 28.79 minutes
- **English:** 96 clips, 22.12 minutes
- **Hindi:** 29 clips, 6.67 minutes
- **Emotions:** neutral (92), narrative (10), excited (8), angry (8), formal (4), happy (3)
- **Validation status:** PASS

Note: Although the original target was approximately 60 minutes, I prioritized validated clean clips and strict audio quality over blindly padding the dataset with noisy or poorly segmented speech to meet the target duration under deadline constraints.

3. Pipeline Overview

The end-to-end dataset generation pipeline consists of the following stages:

1. **YouTube Audio Sourcing:** Selecting clean, high-quality, single-speaker videos.
2. **Audio Download:** Downloading source audio via yt-dlp.
3. **WAV Conversion:** Resampling to 22050 Hz, 16-bit, mono WAV format using ffmpeg.
4. **Segmentation:** Splitting the converted audio into 8-29s clips using pydub silence detection.
5. **Sarvam ASR:** Transcribing the segments using the Sarvam speech-to-text API (saarika:v2.5).
6. **Sarvam Diarization/Speaker Filtering:** Manual speaker validation to ensure single-speaker purity.
7. **Sarvam LLM Emotion Tagging:** Running the zero-shot emotion classifier via the Sarvam 30B LLM.
8. **Manual / Fast QC:** Running automated validations and manually reviewing high-risk and medium-risk clips.
9. **HuggingFace Packaging:** Organizing metadata and audio into a standardized format.

4. Source Selection Criteria

To ensure high-quality speech data, sources were selected based on the following rules:

- **Single-speaker only:** No interviews, debates, or multi-speaker dialogs.
- **Clear audio:** Minimal background noise, no background music, and no crowd noise.
- **No distractions:** Avoided videos with heavy clapping, laugh tracks, or whispering.
- **High-quality speech:** Standard Indian English or clean Hindi lectures with consistent speaking pace.

5. Audio Processing and Segmentation

Source audio was converted to mono WAV at 22050 Hz. Segmentation was performed using pydub's silence-based splitter (`min_silence_len=600ms`, `silence_thresh=audio.dBFS-14`). Any segments with mid-word cuts, overlapping voices, or excessive ambient noise were identified and rejected during the QC phase.

6. Sarvam API Usage

- **Sarvam ASR:** Transcribed all segments using the `saarika:v2.5` model.
- **Sarvam Diarization/Speaker Filtering:** Speaker filtering was handled at the source level. We selected only single-speaker videos, which bypassed multi-speaker diarization and ensured maximum voice consistency.
- **Sarvam LLM:** Evaluated the emotion and style tags using the `sarvam-30b` model.

7. Emotion/Style Tagging

Allowed tags were restricted to the following set: neutral, narrative, excited, angry, formal, happy, sad, calm, whisper. The final distribution is:

- **neutral:** 92 clips (73.6%)
- **narrative:** 10 clips (8.0%)
- **excited:** 8 clips (6.4%)
- **angry:** 8 clips (6.4%)
- **formal:** 4 clips (3.2%)
- **happy:** 3 clips (2.4%)

8. Manual QC and Fast QC

We prioritized transcript correctness and clean audio near the deadline using an automated risk-filtering strategy:

- **High-Risk Filter:** Clips with empty text, length $< 8s$ or $> 45s$, or transcription tokens indicating music/applause.

- **Medium-Risk Filter:** Clips longer than 30s or classified with highly emotional tones.
- **Low-Risk Filter:** Standard short, neutral clips. All high and medium-risk clips were manually listened to and corrected or rejected, while low-risk clips were sampled (every 5th clip) to verify quality. Clips containing cuts, clapping, or music were rejected entirely.

9. Iterations and Improvements

- Fixed encoding issues during Hindi text transcription.
- Switched away from problematic Hindi/Sanskrit-heavy lecture content to standard conversational Hindi.
- Cleaned metadata columns to match standard HuggingFace Hub dataset structures.
- Normalized audio paths to relative format (`audio/filename.wav`).

10. What Worked

- Sarvam ASR provided highly accurate first-pass transcriptions for both English and Hindi.
- Silence-based segmentation effectively split files without cutting mid-sentence.
- Automated validation scripts successfully caught formatting errors before packaging.

11. What Did Not Work

- YouTube rate limits and bot-detection schemes temporarily blocked yt-dlp.
- Some Hindi sources contained Sanskrit slokas which the ASR model struggled to transcribe correctly.
- Hard deadline constraints limited the total volume of Hindi data we could extract and verify.

12. Quality Decisions

- Rejected clips containing mid-word cuts, background noise, and overlapping audio.
- Retained only clips that passed validation.
- Accepted a shorter total duration (28.79 minutes) to maintain a highly reliable, high-fidelity dataset, avoiding low-quality fillers.

13. Final Dataset Links

- **HuggingFace Dataset:** <https://huggingface.co/datasets/champTUSHARg007/indian-tts-dataset>
- **GitHub Repository:** <https://github.com/tusharg007/sarvam-pro>

14. Limitations

- Total duration (28.79 minutes) is lower than the 60-minute target.
- Hindi clips make up only 6.67 minutes of the dataset.

- Emotion tags are LLM-generated and are only approximate.

15. Future Improvements

- Add 3-4 more clean, single-speaker Hindi lecture sources to balance the language representation.
- Run full human QC on all transcribed segments.
- Balance the emotion/style tags across the dataset.